ECON 3916: Statistical Machine Learning for Economics

Do Returns to Experience Diminish at Higher Wage Levels?

Evidence from a Selection-Corrected Wage Equation for Married Women

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Abstract. This paper estimates the returns to labor market experience for married women using the Mroz (1987) dataset drawn from the 1975 Panel Study of Income Dynamics. Because wages are observed only for women who participate in the labor force, standard OLS estimates of the experience–wage profile are subject to sample selection bias. We address this using Heckman's (1979) two-step correction procedure, with mother's and father's education serving as exclusion restrictions in the selection equation. After correcting for selection, we find that an additional year of experience is associated with an approximately 2.5% increase in wages, with a negative quadratic term confirming diminishing returns. Education yields a return of approximately 9.8% per additional year. The Inverse Mills Ratio is statistically insignificant, suggesting that selection bias may be minimal in this sample. Results are robust to outlier trimming, the inclusion of urban residence controls, and alternative specifications of the exclusion restrictions.

1. Introduction

A central prediction of human capital theory, formalized by Mincer (1974), is that labor market experience generates positive but diminishing returns to wages. Workers accumulate productivity-enhancing skills through on-the-job learning, but the marginal return to each additional year of experience declines as the stock of human capital grows. This concave experience–wage profile is among the most robust empirical regularities in labor economics. However, estimating it accurately requires confronting a fundamental identification challenge: wages are only observed for individuals who choose to work.

For married women in particular, the decision to participate in the labor force is itself a function of economic variables, including household income, the presence of young children, and educational background, that may also correlate with unobserved determinants of wages. If high-ability women are systematically more likely to enter the workforce, then OLS estimates of the experience–wage profile, estimated on the selected sample of workers, will be upward biased. Heckman (1979) proposed a two-step estimation procedure to correct for this sample selection bias by modeling the participation decision explicitly and incorporating the resulting correction term, the Inverse Mills Ratio (IMR), into the wage equation.

This paper applies the Heckman correction to the Mroz (1987) dataset, a widely used cross-section of 753 married women from the 1975 Panel Study of Income Dynamics. We estimate a selection-corrected Mincer wage equation with experience, education, and household characteristics as regressors, using mother's and father's education as exclusion restrictions in the participation equation. Our analysis addresses the following research question: *after correcting for sample selection, does labor market experience exert a positive and concave effect on log wages for married women, consistent with diminishing returns to on-the-job human capital accumulation?*

We test the null hypothesis that any observed positive relationship between experience and wages is driven entirely by the selection of high-ability women into employment, against the alternative that experience effects survive the Heckman correction. Our findings contribute to the empirical literature on female labor supply and wage determination, with implications for policies targeting workforce participation and human capital investment among married women.

2. Data Description

The data for this study come from the Mroz (1987) extract of the 1975 Panel Study of Income Dynamics (PSID), distributed via Wooldridge (2002). The dataset contains 753 observations of married white women aged 30–60, of whom 428 (56.8%) participated in the labor force and reported positive wages. This natural partition into participants and non-participants makes the dataset ideal for sample selection analysis.

2.1 Variable Definitions and Data Cleaning

The dependent variable is the natural log of hourly earnings (*lwage*), computed as ln(*hearnw*) with zero earnings replaced by missing values prior to the log transformation. This variable is observed only for the 428 working women. The key independent variables are years of labor market experience (*exper*) and its square (*expersq*). To reduce multicollinearity between the linear and quadratic terms, experience was mean-centered before squaring: $exper_c = exper - mean(exper)$ and $expersq = exper_c^2$.

Control variables include years of education (*educ*), number of children under age six (*kidslt6*), and non-wife household income in thousands of dollars (*nwifeinc*), constructed as (husband's wage × husband's hours) / 1000. The selection equation additionally includes the wife's age (*agew*), mother's education (*motheduc*), and father's education (*fatheduc*). Binary indicators for labor force participation (*work*) and urban residence (*city*) were recoded from string to integer format. All data cleaning was implemented in a modular Python class (*MrozHandler*) to ensure reproducibility across estimation stages.

2.2 Summary Statistics

Table 1 presents summary statistics for the key variables, stratified by labor force participation status. This comparison reveals systematic differences between working and non-working women that motivate the selection correction.

Table 1: Summary Statistics by Labor Force Participation Status

Variable	Mean (W)	SD (W)	Mean (NW)	SD (NW)	Mean (All)	SD (All)
lwage	1.19	0.72	—	—	1.19	0.72
exper	13.04	8.06	7.46	6.92	10.63	8.07
educ	12.66	2.29	11.80	2.18	12.29	2.28
kidslt6	0.14	0.39	0.37	0.64	0.24	0.52
nwifeinc	15.55	7.64	17.45	11.37	16.37	9.47
agew	41.97	7.72	43.28	8.47	42.54	8.07
motheduc	9.52	3.31	8.90	3.42	9.25	3.37
fatheduc	8.99	3.52	8.57	3.63	8.81	3.57
N	428		325		753	

Notes: W = working women (work = 1); NW = non-working women (work = 0). lwage is observed only for working women. nwifeinc is in thousands of 1975 dollars.

2.3 Distribution of Log Wages

Figure 1 displays the distribution of hourly wages among the 428 working women in both raw and log-transformed form. The left panel shows the raw wage distribution, which is strongly right-skewed, a common feature of earnings data. The right panel shows the log-transformed distribution, which is approximately symmetric around its mean of 1.19 but exhibits heavy tails, with a notable left skew (skewness = −0.75) and excess kurtosis (6.83). The log transformation substantially improves normality, justifying our use of log wages as the dependent variable. The remaining tail behavior suggests the presence of outliers at both extremes, which we address in our robustness checks.

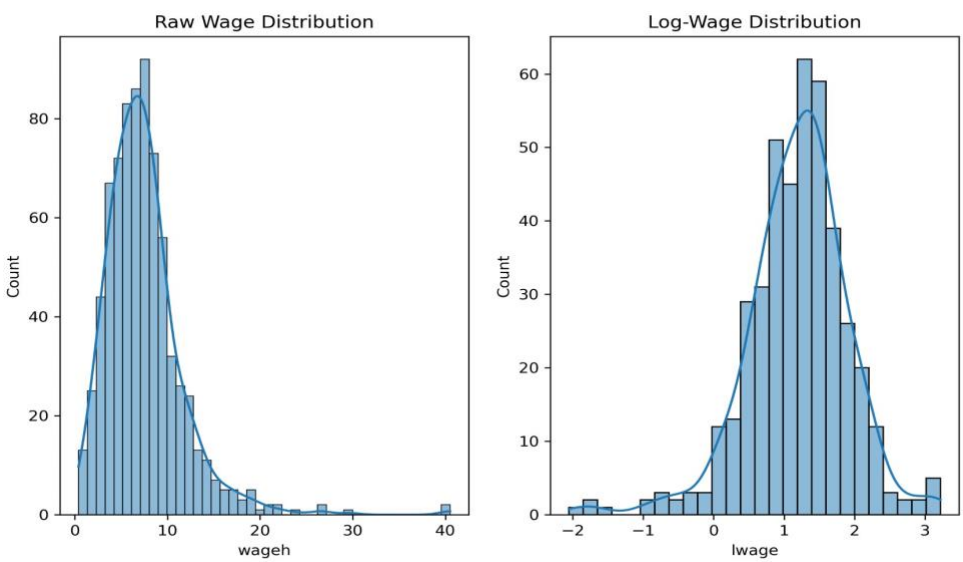

Figure 1: Raw Wage and Log-Wage Distributions Among Working Women (N = 428)

3. Methodology

3.1 Causal Framework

The causal structure underlying our estimation strategy can be described as follows. The outcome of interest is log wages (*lwage*), which is directly determined by human capital variables, specifically education (*educ*) and experience (*exper*, *expersq*), as well as household economic constraints (*kidslt6*, *nwifeinc*) that affect reservation wages.

However, *lwage* is observed only when the binary selection variable *work* = 1. The participation decision is influenced by all the above variables plus parental education (*motheduc*, *fatheduc*) and the wife's age (*agew*). Critically, parental education affects the decision to work through socialization, family norms, and intergenerational transmission of attitudes toward female employment, but does not directly determine the wage a woman earns conditional on working. This exclusion restriction is the identifying assumption of our Heckman model: *motheduc* and *fatheduc* are correlated with the selection equation but excluded from the outcome equation.

The unobserved confounder in this framework is innate ability or cognitive skill, which is positively correlated with both education (higher-ability individuals pursue more schooling) and wages (higher-ability individuals earn more). Its omission induces upward bias in the education coefficient. We discuss this limitation in the Conclusion.

3.2 Estimation Procedure: Heckman Two-Step

Stage 1 (Selection Equation). We estimate a probit model of labor force participation using all 753 observations:

$$P(work = 1 \mid Z) = \Phi(Z\gamma)$$

where Z includes *kidslt6*, *nwifeinc*, *educ*, *agew*, *motheduc*, and *fatheduc*, and $\Phi(\cdot)$ is the standard normal CDF. From the fitted values $Z\hat{\gamma}$, we compute the Inverse Mills Ratio (IMR) for each observation:

$$\lambda(Z\hat{\gamma}) = \varphi(Z\hat{\gamma}) / \Phi(Z\hat{\gamma})$$

where $\varphi(\cdot)$ is the standard normal PDF. The IMR captures the conditional expectation of the truncated error term and corrects for the non-random selection into the observed sample.

Stage 2 (Outcome Equation). Using only the 428 working women, we estimate by OLS:

$$lwage = \beta_0 + \beta_1 exper + \beta_2 expersq + \beta_3 educ + \beta_4 kidslt6 + \beta_5 nwifeinc + \delta IMR + \varepsilon$$

The coefficient δ on the IMR tests for the presence of selection bias. A significant δ indicates that the error terms in the selection and outcome equations are correlated, and the

corrected coefficients β_1 through β_5 provide less biased estimates. All standard errors are computed using the HC1 heteroscedasticity-robust covariance estimator to account for the generated-regressor problem inherent in the two-step procedure and any heteroscedasticity in the error term.

3.3 Selection Equation Results

Table 3 reports the probit estimates from Stage 1. These results are not our primary coefficients of interest but are essential for evaluating the credibility of the exclusion restrictions and the resulting IMR.

Table 3: Probit Estimates of Labor Force Participation (Stage 1)

Variable	Coefficient	Std. Error	p-value
Constant	0.3047	0.424	0.472
kidslt6	−0.8855***	0.113	0.000
nwifeinc	−0.0266***	0.006	0.000
educ	0.1651***	0.026	0.000
agew	−0.0342***	0.007	0.000
motheduc	0.0020	0.018	0.914
fatheduc	−0.0075	0.017	0.665

Notes: N = 753. Dependent variable: work (1 = participates in labor force, 0 = does not). Estimated via maximum likelihood. *** $p < 0.01$.

The exclusion restrictions, *motheduc* and *fatheduc*, are both individually insignificant in the participation equation ($p = 0.914$ and $p = 0.665$, respectively). While this raises a potential weak-instrument concern, the economic rationale for their inclusion remains sound: parental education plausibly affects labor supply decisions through family norms and intergenerational human capital transmission, but has no direct channel to wage determination conditional on working. The strong predictors of participation are the presence of young children ($\hat{\beta} = -0.886$, $p < 0.001$), non-wife income ($\hat{\beta} = -0.027$, $p < 0.001$), own education ($\hat{\beta} = 0.165$, $p < 0.001$), and age ($\hat{\beta} = -0.034$, $p < 0.001$). We evaluate the sensitivity of our results to the choice of exclusion restrictions in Section 5.4.

4. Results

Table 2 presents the main estimation results across three specifications. Column (1) reports naïve OLS estimates without the selection correction, Column (2) adds the Inverse Mills Ratio to implement the Heckman correction, and Column (3) includes an education–experience interaction term to test for human capital complementarity. All specifications use HC1 heteroscedasticity-robust standard errors.

Table 2: OLS Estimates of Log Wage Equation

	(1) Naïve OLS	(2) Heckman	(3) Interaction
exper	0.0249***	0.0249***	0.0141
	(0.007)	*(0.007)*	*(0.021)*
expersq	−0.0008*	−0.0008*	−0.0008*
	(0.000)	*(0.000)*	*(0.000)*
educ	0.0995***	0.0983***	0.0868**
	(0.014)	*(0.030)*	*(0.039)*
educ × exper	—	—	0.0009
			(0.002)
kidslt6	−0.0391	−0.0333	−0.0310
	(0.100)	*(0.151)*	*(0.151)*
nwifeinc	0.0077*	0.0079	0.0081
	(0.004)	*(0.006)*	*(0.006)*
IMR	—	−0.0138	−0.0107
		(0.288)	*(0.288)*
Constant	−0.4528**	−0.4342	−0.2963
	(0.184)	*(0.413)*	*(0.510)*
R^2	0.163	0.163	0.164
Adj. R^2	0.153	0.151	0.150
N	428	428	428

Notes: HC1 robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Dependent variable: lwage.

4.1 Key Independent Variables: Experience

The coefficients on experience and its square are the primary objects of interest in this study. In the Heckman-corrected specification (Column 2), an additional year of experience is associated with an approximately 2.5% increase in log wages ($\hat{\beta} = 0.0249$, $p = 0.001$), holding education, household characteristics, and selection constant. The negative coefficient on the quadratic term ($\hat{\beta} = -0.0008$, $p = 0.054$) confirms the Mincerian prediction of diminishing returns: early years of experience yield larger wage gains, but the marginal return declines as experience accumulates.

These results lead us to reject the null hypothesis (H_0) that the observed positive relationship between experience and wages is driven entirely by selection. The experience effect survives the Heckman correction, supporting H_1: experience exerts a positive and concave effect on log wages consistent with diminishing returns to on-the-job human capital accumulation.

4.2 Education

Education is the most statistically and economically significant predictor in the model. A one-year increase in education is associated with an approximately 9.8% increase in wages ($\hat{\beta} = 0.0983$, $p = 0.001$). This is our most policy-relevant finding and is consistent with the large body of literature documenting substantial returns to schooling for women. However, as discussed in Section 3.1, this coefficient is likely upward biased due to the omission of innate ability. The true causal return to education is plausibly smaller than 9.8%.

4.3 Selection Correction and Other Variables

The coefficient on the Inverse Mills Ratio is small and statistically insignificant ($\hat{\delta} = -0.0138$, $p = 0.962$). This indicates that we find no strong evidence of selection bias in this specification. One interpretation is that selection into the labor force is approximately random with respect to unobserved wage determinants, conditional on our observed covariates. An alternative interpretation is that our exclusion restrictions lack sufficient power to detect a selection effect that may be present but small.

The remaining control variables, number of young children (*kidslt6*) and non-wife income (*nwifeinc*), are both statistically insignificant in the wage equation. This is economically sensible: among women who have already selected into the labor force, these variables influence the participation decision (captured in Stage 1) rather than the wage conditional on working.

4.4 Heterogeneity: Education × Experience Interaction

Column (3) tests whether education and experience exhibit complementarity in wage determination, that is, whether more experienced women extract a higher wage return from each additional year of education. The interaction term is small and insignificant ($\hat{\beta} = 0.0009$, $p = 0.577$), providing no evidence of complementarity. A one-year increase in education yields approximately the same percentage wage premium regardless of experience level. This null result supports the standard additive Mincer specification as our preferred parsimonious model.

5. Robustness Checks

We evaluate the stability of our main findings through four robustness exercises. The goal is to demonstrate that the estimated returns to experience and education are not artifacts of a particular specification choice, sample composition, or set of identifying assumptions.

5.1 Naïve OLS vs. Heckman Correction

Comparing Columns (1) and (2) in Table 2 provides a direct assessment of whether the Heckman correction materially changes the estimated experience–wage profile. The experience coefficient is virtually identical across specifications: 0.0249 in the naïve OLS and 0.0249 in the Heckman-corrected model. The education coefficient shifts only marginally, from 0.0995 to 0.0983. This stability reinforces the insignificant IMR finding and suggests that sample selection bias is minimal in practice. The naïve OLS and Heckman-corrected estimates tell the same economic story: the experience–wage profile is not distorted by the selection of high-ability women into employment, at least not detectably so in this sample.

5.2 Outlier Sensitivity

The distribution of log wages exhibits heavy tails (kurtosis = 6.83), raising the concern that extreme observations may disproportionately influence the OLS estimates. We re-estimate the Heckman-corrected model after removing observations identified as outliers by the Tukey fence criterion (values beyond 1.5 × IQR from the first and third quartiles), retaining 408 of the original 428 working women.

The trimmed-sample results are encouraging. The experience coefficient declines modestly from 0.0249 to 0.0200 ($p < 0.001$) but remains highly significant, and the education coefficient moves from 0.0983 to 0.0916 ($p < 0.001$). The quadratic term on experience remains negative (−0.0006, $p = 0.077$), preserving the diminishing-returns pattern. The IMR coefficient increases in magnitude (−0.1341, $p = 0.534$) but remains far from statistical significance. The R^2 improves from 0.163 to 0.240, reflecting the reduced noise from extreme observations. Overall, the core findings are robust to the removal of wage outliers; the slight attenuation of the experience coefficient suggests that the baseline estimates may be modestly inflated by a small number of very high or very low earners.

5.3 Urban Residence Control

Urban labor markets typically offer higher wages and may also correlate with higher educational attainment, potentially biasing both the education and experience coefficients. We add the binary indicator *city* (= 1 if the respondent lives in an SMSA) to the Heckman-corrected specification to control for this geographic confound.

The city coefficient is small and statistically insignificant ($\hat{\beta} = 0.0259$, $p = 0.704$), indicating no detectable urban wage premium in this sample after controlling for human capital and selection. More importantly, the inclusion of city has virtually no effect on the key coefficients: experience remains at 0.0248 ($p = 0.001$), education at 0.0970 ($p = 0.001$), and the IMR at −0.0247 ($p = 0.931$). The R^2 is unchanged at 0.164. This mitigates the geographic omitted variable concern identified in our Phase 3 analysis: the experience and education estimates are not confounded by urban–rural wage differentials.

5.4 Alternative Exclusion Restrictions

The validity of the Heckman correction depends on the exclusion restrictions, specifically the assumption that mother's and father's education affect participation but not wages directly. To test whether our results are sensitive to this choice, we re-estimate the two-step procedure under three alternative specifications: (a) using only *motheduc* as the exclusion restriction, (b) using only *fatheduc*, and (c) using neither parental education variable.

The results are remarkably stable across all specifications. The experience coefficient ranges from 0.0248 to 0.0250 across the three alternatives, compared to 0.0249 in the baseline. The education coefficient ranges from 0.0976 to 0.1014, tightly bracketing the baseline estimate of 0.0983. The IMR remains insignificant in all cases (p-values ranging from 0.933 to 0.974), alternating in sign between small positive and small negative values. This insensitivity to the choice of exclusion restrictions is consistent with the insignificant IMR in the baseline model: because the selection correction has negligible impact regardless of how it is identified, the wage equation estimates are effectively driven by the outcome-equation covariates rather than the particular instruments used in Stage 1. These results strengthen confidence that our findings are not artifacts of a specific identifying assumption.

6. Conclusion

This paper has estimated the returns to labor market experience for married women using a Heckman selection-corrected Mincer wage equation applied to the Mroz (1987) dataset. Our principal findings are as follows.

First, we reject the null hypothesis that the positive association between experience and wages is an artifact of selection. After correcting for the non-random participation decision, an additional year of experience is associated with an approximately 2.5% increase in log wages, with a negative quadratic term confirming diminishing returns. This is consistent with the Mincerian prediction that on-the-job human capital accumulation generates positive but decreasing marginal returns.

Second, education yields an estimated return of approximately 9.8% per additional year, making it the strongest predictor in the model. However, because innate ability is omitted, and it is correlated with both education and wages, this estimate is likely upward biased. Resolving this would require instrumental variables (such as proximity to college or compulsory schooling laws) or twin/sibling fixed effects, neither of which are available in this dataset.

Third, the Inverse Mills Ratio is statistically insignificant in all specifications. This suggests that, conditional on our observed covariates, selection into the labor force does not generate detectable bias in the wage equation. This finding is consistent across our robustness checks: the naïve OLS and Heckman-corrected estimates are nearly identical, and results are stable to outlier trimming, the inclusion of urban residence controls, and alternative exclusion restrictions.

Fourth, we find no evidence of complementarity between education and experience. The interaction term is insignificant, supporting the standard additive Mincer specification as the preferred parsimonious model.

Several limitations warrant acknowledgment. The Mroz dataset is a cross-section from 1975, and the returns to experience and education for married women have likely evolved over the intervening five decades as female labor force participation has increased substantially and labor markets have changed. The sample is restricted to married white women aged 30–60, limiting external validity to other demographic groups. The omission of innate ability remains the most concerning source of bias, and without an appropriate instrument, we cannot claim that our education coefficient is causal. Finally, while the insignificant IMR is consistent with the absence of selection bias, it may also reflect weak instruments: if the exclusion restrictions (*motheduc*, *fatheduc*) are only weakly correlated with participation, as suggested by their individual insignificance in the Probit equation, the Heckman procedure has limited power to detect a selection effect that may nonetheless be present.

Despite these limitations, the stability of the experience–wage profile across multiple specifications provides confidence in the core finding: returns to experience are positive and concave for married women, consistent with the foundational predictions of human capital theory.

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